

POWERFACTORY

PowerFactory 2024

Technical Reference

Differential Protection

RelBiasidiff, TypBiasidiff

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

Publisher:
DIgSILENT GmbH
Heinrich-Hertz-Straße 9
72810 Gomaringen / Germany
Tel.: +49 (0) 7072-9168-0
Fax: +49 (0) 7072-9168-88
info@digsilent.de

Please visit our homepage at:
https://www.digsilent.de

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1 General Description

This document describes different configurations and the corresponding dynamic and steady-state behaviour of the *Differential Protection* unit for protection devices in *PowerFactory*.

1.1 Differential Protection Type (*TypBiasidiff*)

The *Differential Protection Type* defines the general characteristic and available ranges for the differential protection.

1.1.1 Basic Data

The *Type* selection defines the number of CT inputs and the type of differential protection modelled. The following options are available:

- *3ph*: stabilised current comparison with three phases per CT input
- *1ph*: stabilised current comparison with one phase per CT input
- *3ph Phase Comparison*: phase comparison with with three phases per CT input
- *1ph Phase Comparison*: phase comparison with with one phase per CT input
- *Restricted Earth Fault*: stabilised current comparison with one phase per CT input

For stabilised current comparison types, the *Characteristic* defines the used differential characteristic. Please refer to Section 2 for a detailed description. The *Restraint type* defines the method used to calculate the restraint quantity. Please refer to Section 3 for a detailed description.

For the phase comparison types the characteristic described in Section 2.6 is used.

1.1.2 Tap

The *Tap* page contains ranges for tap-based inputs, commonly found in SEL devices. Please refer to Section 1.2 for a detailed description.

1.1.3 Harmonic Blocking

Depending on the application the differential protection can harmonic blocking based on 2nd, 4th and 5th harmonic order. If any of the three *Threshold Ranges* (e:rH2threshold, e:rH4threshold or e:rH5threshold) is entered, harmonic blocking is enabled.

For 3-phase differential types a phase interlock for the harmonic blocking can be defined per harmonic order. The following interlock modes are available.

- *1 out of 3*: All phases are blocked if at least one phase exceeds the threshold;
- *2 out of 3*: All phases are blocked if at least two phases exceed the threshold;

- *Average*: All phases are blocked if the average of the three harmonic ratios exceeds the threshold.

The configuration supports multi-selection. If more than one interlock mode is configured, the phase interlock becomes user-configurable in the element. If no option or only the *Disabled* option is selected, the phase interlock is disabled.

If the option *Harmonic blocking can be disabled* (e:ihblckdisav) is enabled, the harmonic blocking becomes user-configurable in the element.

If the *Disable Harmonic Blocking Range* is entered harmonic unblocking is enabled.

1.2 Differential Protection (*RelBiasidiff*)

Depending on the selection made in the type for a particular setting range, the following unit conversions apply:

$$I_{p.u.} = \frac{I_{sec.A}}{I_{nom}}$$
$$I_{tap} = \frac{I_{sec.A}}{tap}$$
$$I_{I_{diff}} = \frac{I_{sec.A}}{I_{diff}}$$

where:

$I_{p.u.}$ is the setting in p.u.

$I_{sec.A}$ is the setting in sec.A

I_{tap} is the setting in multiples of tap

$I_{I_{diff}}$ is the setting in multiples of the differential threshold (not available for the *Release Threshold* itself)

I_{nom} is the nominal secondary current in sec.A (*e:Inom* of the connected measurement)

tap is the set tap value in sec.A. The value is taken from either *Tap 1* (e:tap1) on the *Taps* page or from *Differential Current* (e:Itapr) on the *Basic Data* page, depending on whichever range is non-empty. If both ranges are provided, *Tap 1* (e:tap1) is used.

I_{diff} is the set *Release Threshold* in sec.A (e:Idiffr)

2 Differential Characteristics

The following sections provide an overview for different characteristics available for the differential protection. For simplicity, parameter explanations use p.u. values. Please refer to Section 1.2 for the transformation into other input units.

2.1 Standard

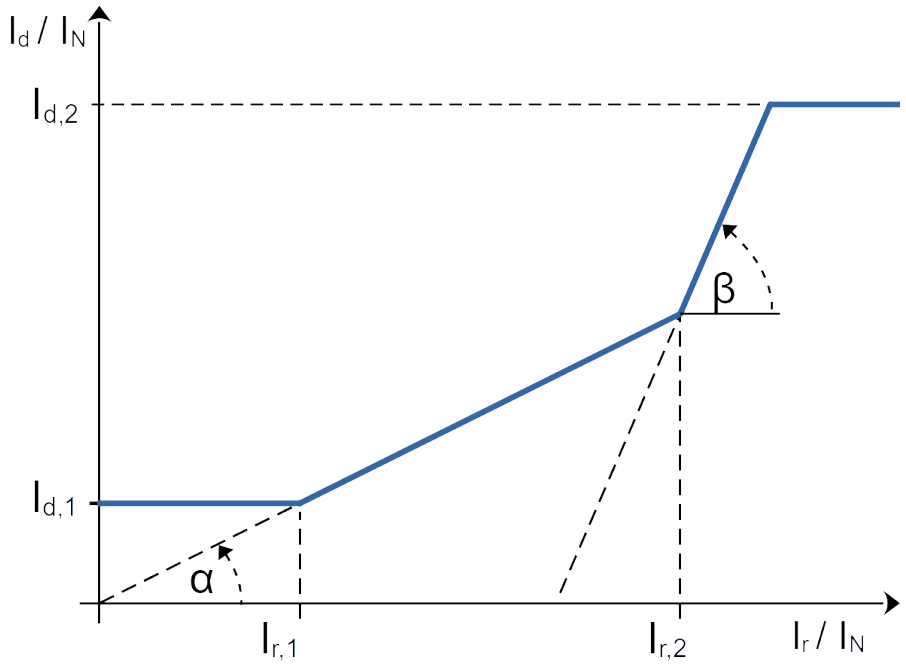


Figure 2.1: Current Comparison Characteristic - *Standard*

The values indicated in Figure 2.1 are defined as follows:

$I_{d,1}$ Differential Release in p.u. (*e:Idiff*)

$I_{d,2}$ Unrestrained Differential Release in p.u. (*e:Idiffunrest*)

$I_{r,1}$ 1st Slope Threshold in p.u. (*e:Ipset1*)

$I_{r,2}$ 2nd Slope Threshold in p.u. (*e:Ipset2*)

α 1st Slope in % (*e:Irestrpercent1*)

β 2nd Slope in % (*e:Irestrpercent2*)

2.2 Independent 2nd Slope

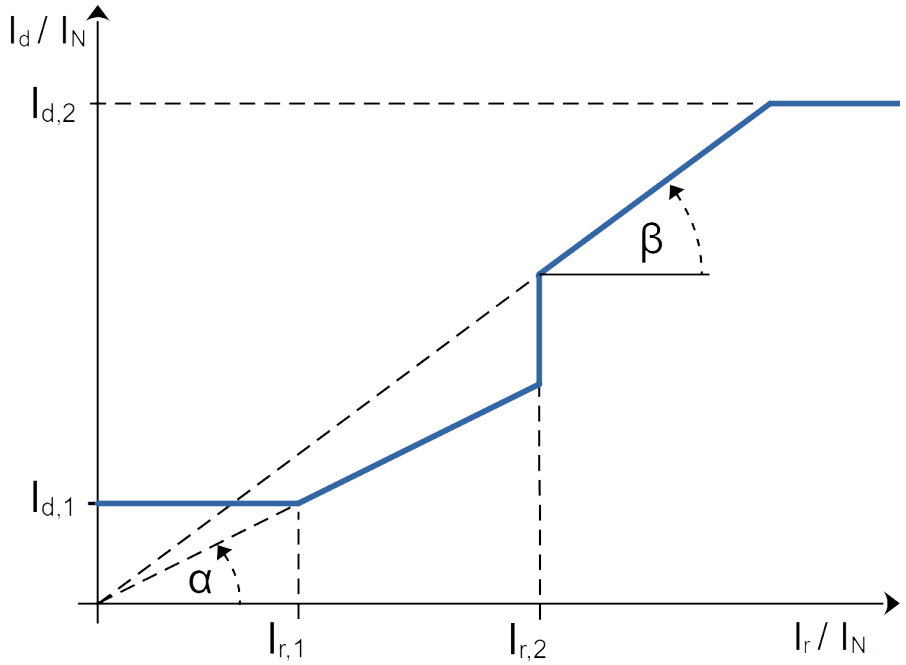


Figure 2.2: Current Comparison Characteristic - *Independent 2nd Slope*

The values indicated in Figure 2.2 are defined as follows:

$I_{d,1}$ Differential Release in p.u. (*e:Idiff*)

$I_{d,2}$ Unrestrained Differential Release in p.u. (*e:Idiffunrest*)

$I_{r,1}$ 1st Slope Threshold in p.u. (*e:Ipset1*)

$I_{r,2}$ 2nd Slope Threshold in p.u. (*e:Ipset2*)

α 1st Slope in % (*e:Irestrpercent1*)

β 2nd Slope in % (*e:Irestrpercent2*)

2.3 All Slopes Independent

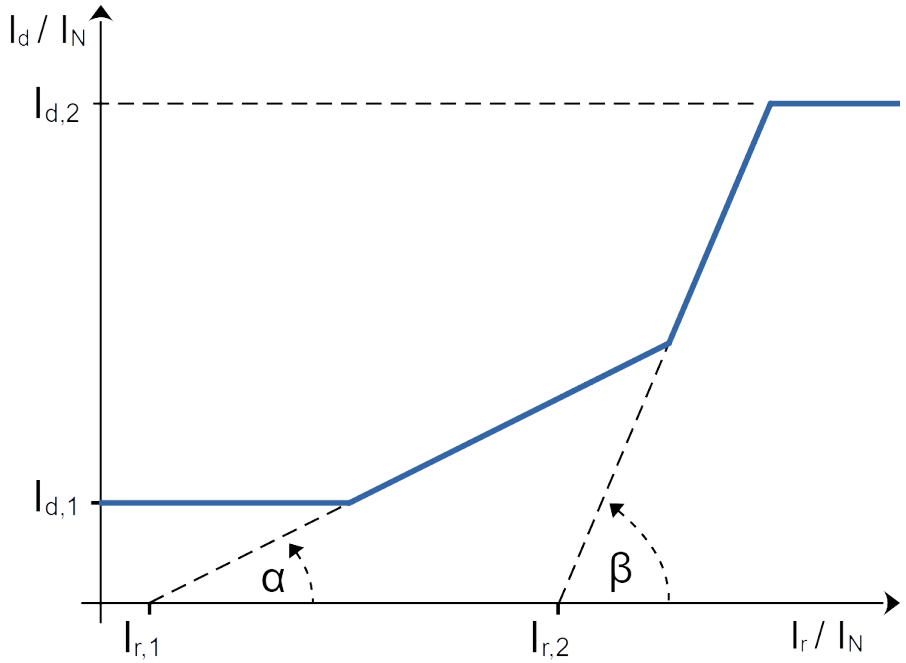


Figure 2.3: Current Comparison Characteristic - *All Slopes Independent*

The values indicated in Figure 2.3 are defined as follows:

$I_{d,1}$ Differential Release in p.u. (*e:Idiff*)

$I_{d,2}$ Unrestrained Differential Release in p.u. (*e:Idiffunrest*)

$I_{r,1}$ 1st Slope Threshold in p.u. (*e:Ipset1*)

$I_{r,2}$ 2nd Slope Threshold in p.u. (*e:Ipset2*)

α 1st Slope in % (*e:Irestrpercent1*)

β 2nd Slope in % (*e:Irestrpercent2*)

2.4 Transition Zone

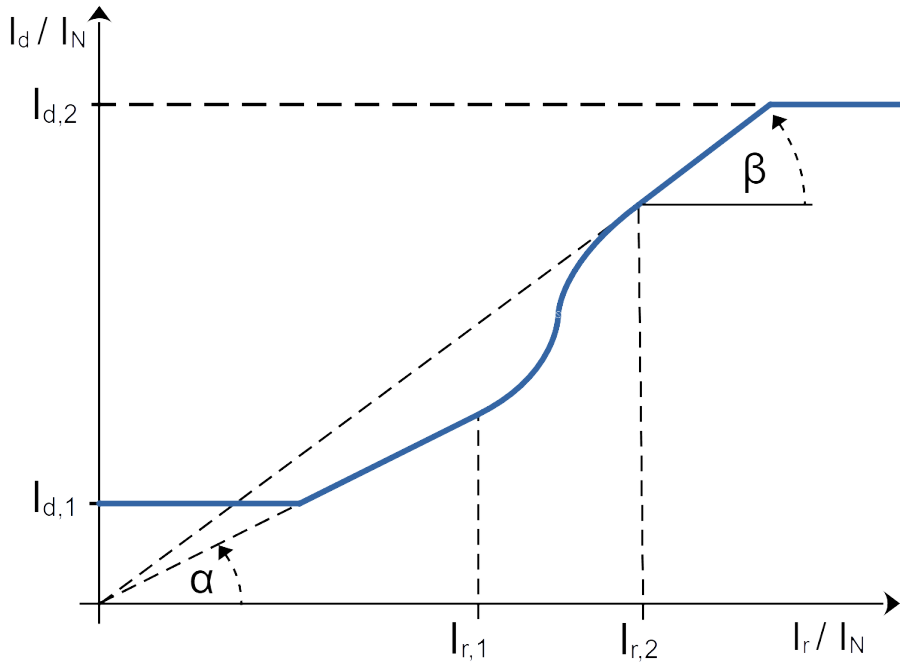


Figure 2.4: Current Comparison Characteristic - *Transition Zone*

The values indicated in Figure 2.4 are defined as follows:

$I_{d,1}$ Differential Release in p.u. (*e:Idiff*)

$I_{d,2}$ Unrestrained Differential Release in p.u. (*e:Idiffunrest*)

$I_{r,1}$ 1st Slope Threshold in p.u. (*e:Ipset1*)

$I_{r,2}$ 2nd Slope Threshold in p.u. (*e:Ipset2*)

α 1st Slope in % (*e:Irestrpercent1*)

β 2nd Slope in % (*e:Irestrpercent2*)

The transition zone between $I_{r,1}$ and $I_{r,2}$ is determined with a Hermite interpolation, configured so that the slopes on either end of the transition zone match.

2.5 Toshiba

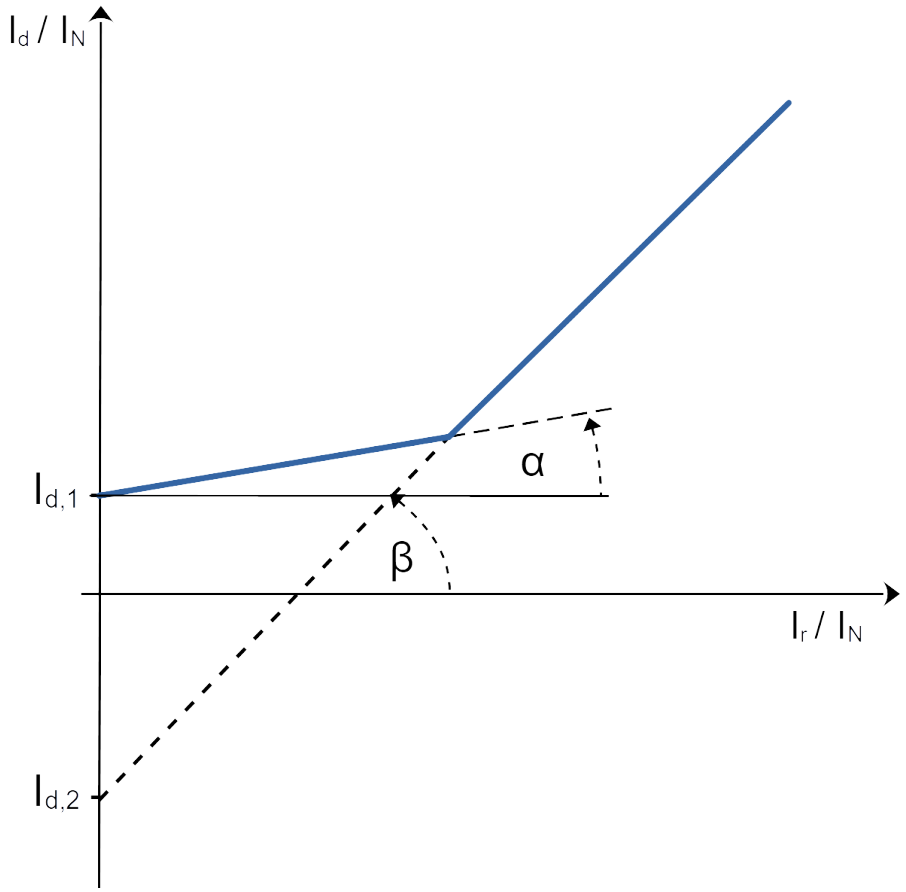


Figure 2.5: Current Comparison Characteristic - *Toshiba*

The values indicated in Figure 2.5 are defined as follows:

$I_{d,1}$ 1st slope threshold

$$I_{d,1} = \frac{5}{6} \cdot \text{DIFI1}$$

with DIFI1 in p.u. (*e:DIFI1*)

$I_{d,2}$ 2nd slope threshold

$$I_{d,2} = -2 \cdot \text{DIFI2}$$

with DIFI2 in p.u. (*e:DIFI2*)

α Fixed 1st slope $\alpha = \frac{1}{6} \cdot 100$ %

β Fixed 2nd slope $\beta = 100$ %

Note: *Stabilised current comparison with one phase per CT input* only uses the first slope when configured as *Toshiba*.

2.6 Phase Comparison

In contrast to other characteristics covered in this section the phase comparison characteristic is unit-less and shown in the complex plane.

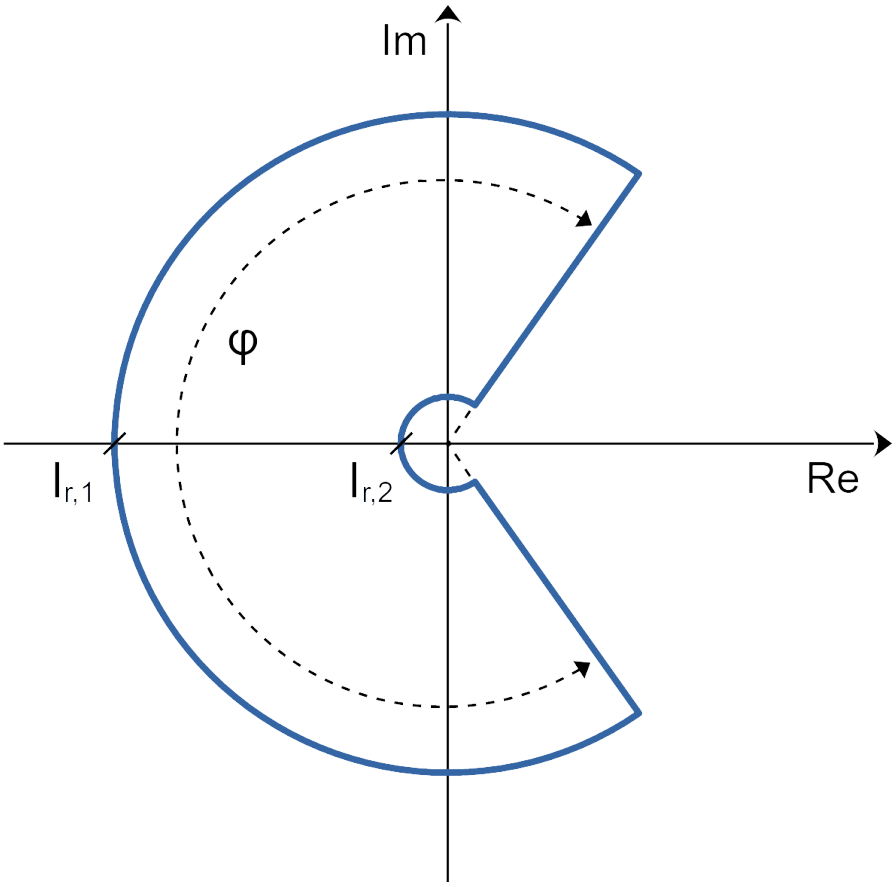


Figure 2.6: Phase Comparison Characteristic

The values indicated in Figure 2.6 are defined as follows:

$I_{r,1}$ Restraint Region Radius (*e:RestrRadius*)

$I_{r,2}$ Inner restraint region radius

$$I_{r,2} = \frac{1}{I_{r,1}}$$

φ Restraint Region Angle in (*e:RestrAngle*)

3 Calculation Models

This section describes the calculation models and possible connection schemes for a differential protection unit in *PowerFactory*. Figure 3.1 shows the schematic connection assumed in the equations.

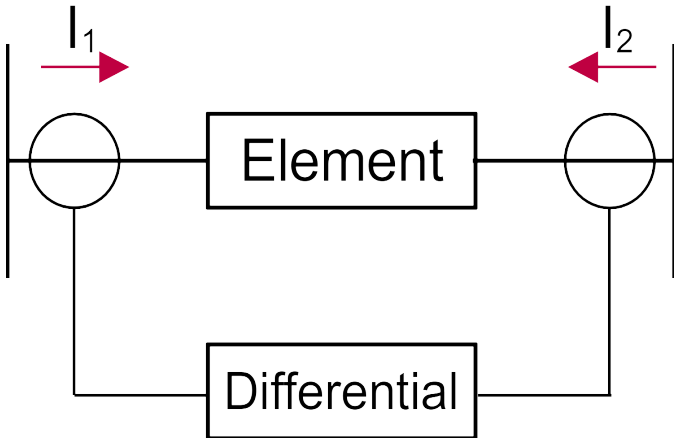


Figure 3.1: Schematic connection of a differential protection

For readability, the equations below are simplified to single phase and currents are expressed as complex phasors. For 3-phase differential protection separate I_d and I_r calculations are performed per phase and the input and output signals receive the corresponding phase suffix ("_A", "_B" and "_C"). Please refer to Section 3.3 for differential protection using instantaneous values.

3.1 Stabilised Current Comparison

Stabilised current comparison calculates the differential and restraint currents and compare the resulting working point against the configured characteristic. If the point is above the characteristic the internal timer starts. Once the timer expires the tripping output activates.

The differential current I_d is calculated as follows:

$$I_d = |\underline{I}_d| = \left| \sum_{i=}^{N} \underline{I}_{CTi} \right| \tag{1}$$

where:

I_d is the magnitude of the differential current in sec.A (*s:Idiffabs*)

$\underline{I}_d$ is the complex differential current in sec.A (*s:Irdiff* $+ j \cdot$ *s:Iidiff*)

$\underline{I}_{CTi}$ is the complex current from CTi in sec.A (*s:IrCTi* $+ j \cdot$ *s:IiCTi*)

The restraint current I_r is calculated as follows if the corresponding input signal (*s:Istababs*) is not connected, depending on the *Restraint Type* selected in the type.

- *Average:*

$$I_r = \sum_{i=1}^{N} \frac{I_{CTi}}{N} \tag{2}$$

- *Scalar Product:*

$$\varphi_{1,2} = \angle\left(\underline{I}_{CT1}; -\underline{I}_{CT2}\right)$$
$$I_r = \begin{cases} \sqrt{|\underline{I}_{CT1}| \cdot |\underline{I}_{CT2}| \cdot \cos\varphi_{1,2}} & \text{if} \quad \cos\varphi_{1,2} \geq 0 \\ 0 & \text{if} \quad \cos\varphi_{1,2} < 0 \end{cases} \tag{3}$$

- *Maximum:*

$$I_r = \max\left(I_{CT1}; \ldots; I_{CTN}\right) \tag{4}$$

- *Scalar Sum:*

$$I_r = \sum_{i=1}^{N} I_{CTi} \tag{5}$$

where:

I_r is the magnitude of the restraint current in sec.A

I_{CTi} is the current magnitude from CTi in sec.A (s:ICTi)

$\underline{I}_{CTi}$ is the complex current from CTi in sec.A (*s:IrCTi* $+ j \cdot$ *s:IiCTi*)

Note: Due to the specialised equation, the *Scalar Product* restraint calculation can only be used with two CTs connected.

3.2 Phase Comparison Differential

Phase comparison differential calculates a complex quotient $\underline{k}$ from the two inputs and compares it against the characteristic. If the quotient lies outside the characteristic the internal timer starts. Once the timer expires the tripping output activates.

The complex quotient $\underline{k}$ is calculated as follows:

$$\underline{k} = \frac{\underline{I}_{CT1}}{\underline{I}_{CT2}} \tag{6}$$

where:

$\underline{k}$ is the complex quotient

$\underline{I}_{CTi}$ is the complex current from CTi in sec.A (*s:IrCTi* $+ j \cdot$ *s:IiCTi*)

3.3 Connection Schemes

All equations in this document are expressed with complex phasors, as almost all calculations, except the EMT simulations, in *PowerFactory* use complex phasors. In most cases, differential protection models will also use complex phasors in EMT simulations, due to the digital filtering performed by the measurement units. This section illustrates possible signal connections for differential protection depending on the measurement configuration.

For simplicity, only directly connected measurements are used here. Especially for transformer differential protection, a CT Adapter (*RelCtadapt*) may be interposed between measurement and differential protection. Please refer to the Technical Reference of the adapter for more information.

It is recommended to use either digital or analogue measurements for all connections. Please refer to the Techical Reference of the Measurement (*RelMeasure*) for detailed information about the configuration of the measurement.

3.3.1 Digital Measurement

When using measurements with digital filtering enabled (e.g DFT), the output of the measurement and therefore the input of the differential protection are complex phasors. This enables direct calculation of all required quantities, as indicated throughout Section 3. The simplest possible connection scheme is shown in in Figure 3.2

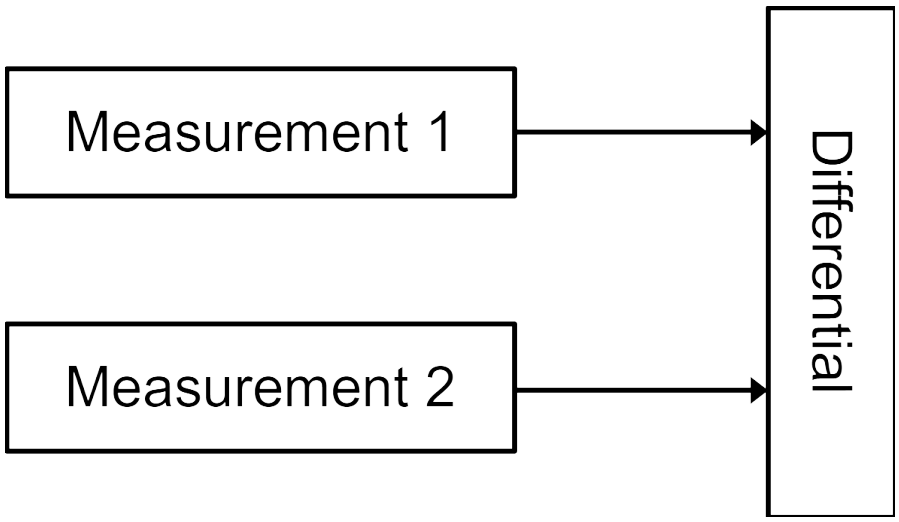


Figure 3.2: Example connection for digital measurements

With digital measurements it is sufficient to connect the signals for magnitude, real, and imaginary part (*s:ICTi*, *s:IrCTi*, *s:IiCTi*). Differential and restraint current will be calculated internally and used for the calculation.

It is possible to bypass the internal calculation by connecting the input signals for differential (*s:Idiffabs*) and/or (*s:Istababs*). The input signals always take precedence over the internal calculation.

Note: *Phase Comparison Differential* units or units with the *Restraint Type* set to “Scalar Product” must be connected to digital measurements to ensure correct operation.

3.3.2 Analogue Measurement

When using measurements without digital filtering enabled, the input for the differential protection consists of instantaneous values. The general equations can still be used, but an additional measurement feedback loop is required. This measurement ensures a correct magnitude calculation for the differential current, which would oscillate otherwise.

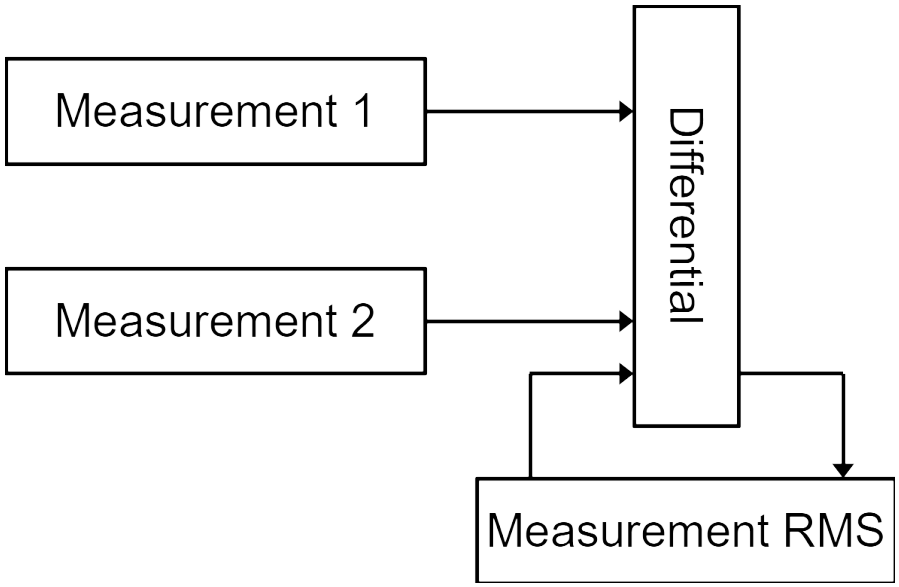


Figure 3.3: Example connection for analogue measurements

With analogue measurements the signals for magnitude and real part must be connected (*s:ICTi*, *s:IrCTi*). In addition, the differential output signal (*s:Irdiff*) must be connected to a suitably configured measurement. Its output must connected to the differential current magnitude input (*s:Idiffabs*), closing the feedback loop.

The restraint current calculation can be used as is, as it only uses the magnitudes.

3.3.3 Harmonic Blocking

Harmonic blocking is only available for EMT simulations, as other calculations only consider the fundamental frequency. Depending on the connected harmonic signals, different variants of the harmonic ratio calculation can be achieved. The examples below use the signals for the 2nd harmonic for readability. The connections for 4th and 5th harmonic are analogous.

To calculate the harmonic ratio based on the harmonic content of the differential current, connect the *s:Harm2Idiff* signal. This signal can be calculated similarly to the differential current shown in Figure 3.3. The ratio is calculated as follows:

$$ratio = I_{d,h}/I_d \tag{7}$$

For digital measurements, it is sufficient to connect the real and imaginary parts for the the 2nd harmonic of the individual CTs (*s:Harm2IrCTi* and *s:Harm2IiCTi*). The harmonic differential current will be calculated from the inputs and used in the equation.

To base the harmonic ratio on the fundamental frequency value of the differential current, the corresponding inputs for the fundamental frequency (*s:Harm1Idiff* for analogue measurements

or *s:Harm1IrCTi* and *s:Harm1IiCTi* for digital measurements) need to be connected as well. The equation then becomes:

$$ratio = I_{d,h}/I_{d,1} \tag{8}$$

To calculate the harmonic ratio based on the harmonic content of individual CT connections, only the harmonic magnitude signals for the individual CTs must be connected (*s:Harm2ICTi*). The ratio is calculated as follows:

$$ratio = \max(I_{CT1,h}/I_{CT1}; \ldots; I_{CTN,h}/I_{CTN}) \tag{9}$$

To base the harmonic ratio on the fundamental frequency value for the individual CTs, connect the fundamental frequency magnitude signal as well (*s:Harm1ICTi*). The equation then becomes:

$$ratio = \max(I_{CT1,h}/I_{CT1,1}; \ldots; I_{CTN,h}/I_{CTN,1}) \tag{10}$$

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